

Lecture – 01 Introduction To JavaScript

The Frontend Triad: Core Difference

- **HTML:** Structure of a webpage (Creates a **Static Website**).
- **CSS:** Presentation and layout (Adds **Design and Styling**).
- **JavaScript:** Behaviour and logic (Makes the site **Interactive and Dynamic**).

What is JavaScript?

JavaScript is a high-level, lightweight programming language used to add interactivity, logic, and dynamic behaviour to web pages.

Example: Clickable buttons, Search functionality, etc.

Why Does JavaScript Exist?

The Problem: In 1995, **Netscape Navigator** was a purely static browser. There was no programming language running on the client-side (inside the browser). If a user filled out a form, the input fields could only be validated by sending the data across the network to the server. If a single field was wrong, the server had to send the entire page back, creating massive delays.

The Story of Netscape, Sun Microsystems and Microsoft:

During this era, tech dominance looked like this:

- **Netscape:** Owned the market-leading web browser.
- **Sun Microsystems:** Owned **Java**, a highly popular language running on physical hardware.
- **Microsoft:** Owned the **Windows Operating System** and held massive control over the software market.

The Alliance against Microsoft: Netscape partnered up with Sun Microsystems. Their revolutionary goal was to allow complex applications to run **directly inside the browser**. If a user could do everything inside a web browser, the underlying operating system (Windows) would become irrelevant. This triggered the famous "Browser Wars" of the late 1990s.

Sun Microsystems announced that Java could run on both the backend and the frontend. Netscape initially wanted to integrate Java directly into the browser. However, they realized Java was too complex and heavy. They needed a small, lightweight companion "scripting language" instead.

The Creation of JavaScript:

Netscape hired **Brendan Eich** to integrate a language called Scheme into the browser. However, due to the alliance with Sun, management ordered him to make the new language look syntactically similar to Java to ride the wave of Java's popularity (even though under the hood, Java and JavaScript are fundamentally distinct).

Driven by aggressive deadlines to beat Microsoft's upcoming browser, Brendan Eich famously designed and coded the first version of the language in just **10 days** in May 1995 (initially code-named Mocha, later renamed LiveScript, and finally marketed as JavaScript).

Why a New Language?

1. Easy to Learn for Web Developers

JavaScript is easy to learn for developers who already know HTML and CSS because:

- It follows a simple syntax
- It integrates directly with HTML
- No complex setup is required

2. Security Limitations

JavaScript was designed with security restrictions to protect users, such as:

- No direct access to the user's file system
- Runs inside a browser sandbox
- Prevents harmful system-level operations

These limitations help ensure safe execution in web browsers.

3. Lightweight Design

JavaScript is a lightweight language because:

- It does not require heavy compilation
- Executes quickly in the browser

This makes web pages faster and more responsive.

Client-Server Interaction:



Browser can read, understand and execute JavaScript.

Why C++ Wasn't Used?

- **Security:** C++ has low-level, direct access to a computer's system memory and hardware. Running unverified C++ code from the internet would allow malicious websites to easily corrupt or steal data from the host machine.
- **Complexity:** C++ is a heavy, complex language that requires manual memory management and compilation.

Why Java Applets Were Discontinued?

Java Applets allowed full Java applications to run inside a browser plugin. However, they were heavyweight, slow to load, required a massive plug-in architecture, and suffered from severe, recurring security flaws.

Note: Modern browsers can run low-level languages like C++, C#, and Rust at near-native speeds using WebAssembly (Wasm).

What is WebAssembly (Wasm)?

WebAssembly is a binary instruction format that runs inside a highly secure, sandboxed browser environment, allowing heavy computation directly in the web browser without native security risks.

How C++ Code Runs?

C++ is a compiled language. It is translated completely into machine code before execution.

C++ Code → Compiler → Machine Code → CPU → Output

How JavaScript Runs inside the Browsers?

JavaScript runs inside the browser using a JavaScript engine.

- Google Chrome & Microsoft Edge: V8 Engine
- Mozilla Firefox: SpiderMonkey
- Apple Safari: JavaScriptCore

JavaScript Code → V8 Engine → Output

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